

ABSTRACT

These days, we find many batteries used in Electric Vehicles are burning or blasting due to the overcharging or long time charging. During such condition, the battery body temperature may rise by which the batteries are blasting. To avoid these types of mishaps, here this special type of battery charger is designed using latest technology such that the battery can be charged using wireless technology and at the same time the battery voltage and its body temperature data will be monitored continuously through an embedded system. An important feature added in the system is that if the battery body temperature raises more than the threshold value, immediately supply to the battery will disconnected automatically and alarm will be energized. Once the alarm is energized it remains in energized condition until the reset button is activated.

The battery condition will be monitored digitally and it will be displayed through an LCD interfaced with Arduino board. Another important feature added to the system is that entire information will be transmitted to the concerned mobile phone through WiFi module using IOT technology. The data will be transmitted to the phone directly because the wireless link will be established between the system and mobile phone. Wireless charging is a new technology for charging the batteries that allows charging over short distances without cables. The advantage of wireless charging is that charging is quicker and easier, we need not have to plug and unplug each time, simply by placing the vehicle in its parking place where the power transmitting coil is installed under the ground, the battery starts charging automatically. For demo purpose, 4 wheelers chassis will be constructed and all devices including battery will be arranged over this manually moving body. The power receiver coil will be arranged below the chassis such that it acquires maximum current from the wireless charger.

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LIST OF SYMBOLS

ABBREVIATION	NAMES
A	Ampere
V	Voltage
PCB	Printed Circuit Board
IC	Integrated Circuit
R	Resistor
C	Capacitor
I	Current

CHAPTER - 1

INTRODUCTION

1.1 INTRODUCTION

In this chapter, we will discuss about the main objective of the project.

A wireless charger for electric vehicles is a new subject for the discussion. Many scientists across the world are conducting many experiments over this technology to improve the efficiency and distance between the power transmitting coil and power receiving coil. Presently the main drawback of this system is, poor efficiency and poor distance, once this is improved this technology will become popular and we can find these chargers everywhere. In practical, in this method of charging, when considered to charge an electric car battery, the power receiving coil must be arranged below the chassis of vehicle. Here vehicle is not constructed, but demo piece can be simulated. With the help of a digital monitoring system constructed with Arduino board, battery body temperature and its voltage can be monitored during charging or in idle condition. 12v – 2Ah rechargeable battery is used and its terminal voltage is monitored continuously through an LCD interfaced with the Arduino board.

When the EV batteries are charging in hot environment especially in summer, battery cathodes are main cause of the heat release. Protecting the battery with efficient thermal management is essential, especially in our country, the ambient temperature is high. Therefore this project work is considered to protect the battery burning due to the over temperature during charging.

In wireless chargers, often a power transmitting coil as well as power receiving coil plays dominant role for charging the batteries. In this method, the energy will be transmitted to one coil to the another coil without any conducting wires. Resonant inductive coupling method is adopted which is also known as electro dynamic induction. Wireless power techniques are two categories, near field and far field. Here we used near field or non-radiative techniques, power is transferred over short distances by magnetic fields using inducting coupling between coils of wire, or by electric fields using capacitive coupling between metal electrodes. Inductive coupling is most widely used wireless technology. When a secondary coil is brought near to this magnetic field, maximum energy will be grabbed which is converted as pure DC source and which can be used to charge the battery. The output is no regulated there by voltage up and downs can be monitored by varying the distance between two coils. The Arduino processor

used in the project work is having built-in-with ADC, so that additional ADC is not required for converting the analog data generated by the temperature sensor. In same manner, the battery terminal voltage is also monitored and displayed through an LCD interfaced with Arduino.

OBJECTIVE

1.2 OBJECTIVE

The objective of the project "Hi-tech battery charger for modern EVs using IOT" is to develop a smart battery charging system that can efficiently and safely charge the batteries of modern electric vehicles (EVs). The system will leverage the power of the Internet of Things to enable remote monitoring, control, and management of the charging process. The main measures to improve the wireless charging efficiency of electric vehicles and the electromagnetic interference and radiation problems in the process of wireless charging are analyzed. Charging of an electric vehicle can be performed by either conductive (or wired) charging or wireless charging. Wired charging uses connection means between electric supply and charge inlet of the vehicle. Even though wired charging is popular, the problems with messy wires and safety matter in wet environment are a major drawback of this charging. Since a few years, a large interest is growing for the supply of the electric loads through a field to dispense from any wired connection with the grid. The apparatuses that actuate the through the-field supply are termed as wireless power transfer systems (WPTSs). Their deployment has started for the recharge of the batteries that are fitted in grid-detachable equipment. Recharging is executed while the equipment is standing in an on-purpose set but the longtime perspective is the supply of equipment while moving, with the purpose of removing the batteries or at least of reducing their capacity. Wireless charging of the EV batteries offers a number of advantages compared to the wired counterpart; indeed, wireless charging makes it i) unnecessary any plug, cable or outlet, ii) friendly the charging process, iii) fearless the transfer of energy in any environmental condition, and so on. For these reasons, WPTSs are expected to play a major role in the future charging process for the EVs. Finally the main problems and solutions of the wireless charging technology of electric vehicles are studied. Electrified transportation will minimize greenhouse gas emissions while also lowering gasoline prices. To encourage the adoption of electrified transportation, a variety of charging networks must be established in a user-friendly environment. Wireless electric vehicle charging systems could be a viable

alternative technology for charging electric vehicles without the need for a plug. The work done in the area of wireless power transfer technology for electric vehicles is the main objective of our project

1.3 RELATED WORKS

In this project, we have developed an important feature added in the system is that if the battery body temperature rises more than the threshold value, immediately supply to the battery will disconnect automatically and alarm will be energized. Once the alarm is energized it remains in energized condition until the reset button is activated. The battery condition will be monitored digitally and it will be displayed through an LCD interfaced with Arduino board. Another important feature added to the system is that entire information will be transmitted to the concern mobile phone through WIFI module using IOT technology. The data will be transmitted to the phone directly because the wireless link will be established between the system and mobile phone. Wireless charging is a new technology for charging the batteries that allows charging over short distances without cables. The advantage of wireless charging is that charging is quicker and easier, we need not have to plug and unplug each time, simply by placing the vehicle in its parking place where the power transmitting coil is installed under the ground, the battery starts charging automatically.

1.4 ORGANISATION OF THESIS

CHAPTER 1:

In this chapter, we will discuss about the main objective of the project.

CHAPTER 2:

In this chapter, we will discuss about the components of the project.

CHAPTER 3:

In this chapter, we will discuss about the operation of the project.

CHAPTER 4:

In this chapter, we will discuss about the result of the project.

CHAPTER 5:

In this chapter, we will get the conclusion of Hi-Tech battery charger for modern EV's using IOT.

1.5 CONCLUSION

In this chapter, we made a brief discussion on the introduction and objective of project.

CHAPTER – 2

COMPONENTS

2.1 INTRODUCTION

In this chapter, we will discuss about the components of the project.

MAJOR COMPONENTS IN THE PROJECT

2.2 ARDUINO PROCESSOR

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are used for multiple applications in the field of electronics and communications. Arduino is a basic single board microcontroller designed to make applications, interactive controls, or environments easily adaptive. The hardware consists of a board designed around an 8-bit microcontroller, or a 32-bit ARM. Current models feature things like a USB interface, analog inputs, and GPIO pins which allow the user to attach additional boards. GPIO stands for General Purpose, Input, and Output. All the processors we use to have at least a few, a Raspberry Pi and an Arduino have a lot of General-Purpose Input Output that we can design our circuits.

2.2.1 HISTORY OF ARDUINO

Introduced in 2005, the Arduino platform was designed to provide a cheaper way for hobbyists, students and professionals to create applications that play in the human interface world using sensors, actuators, motors, and other elementary products. Common applications for students or the inexperienced are simple robots or motion detectors. It offers a simple integrated IDE (integrated development environment) that runs on regular personal computers and allows users to write programs for Arduino using C or C++.

Typical prices of Arduino boards run around Rs; 300/-. Arduino boards can be purchased pre-assembled or as do-it-yourself kits. Hardware design information is available for those who would like to assemble an Arduino by hand. Over the years Arduino has been the brain of thousands of projects, from everyday objects to complex scientific instruments. A worldwide community of makers - students, hobbyists, artists, programmers, and professionals - has gathered around this open-source platform, their contributions have added up to an incredible amount of accessible knowledge that can be of great help to novices and experts alike

Arduino was born at the Ivrea Interaction Design Institute, Italy as an easy tool for fast prototyping, aimed at students without a background in electronics and programming.

2.2.2 WHY ARDUINO

Thanks to its simple and accessible user experience, Arduino has been used in thousands of different projects and applications. The Arduino software is easy-to-use for beginners, yet flexible enough for advanced users. It runs on Mac, Windows, and Linux. Teachers and students use it to build low cost scientific instruments, to prove chemistry and physics principles, or to get started with programming and robotics. Designers and architects build interactive prototypes, musicians and artists use it for installations and to experiment with new musical instruments. Makers, of course, use it to build many of the projects exhibited at the Maker Faire, for example. Arduino is a key tool to learn new things.

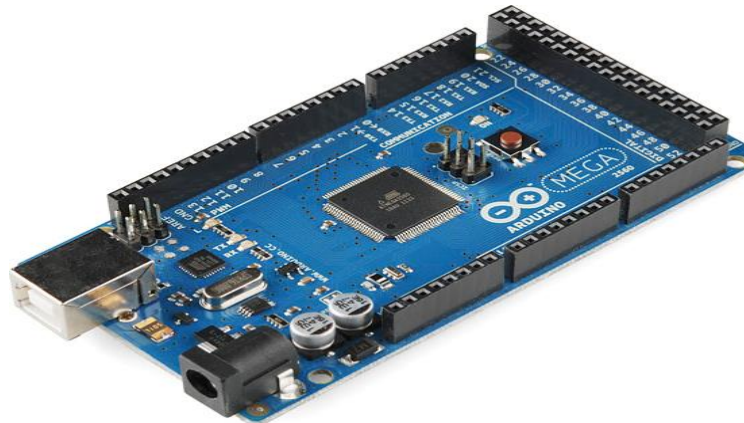


Fig 2.1 Arduino Microprocessor

There are many other microcontrollers and microcontroller platforms available for physical computing. Parallax Basic Stamp, Netmedia's BX-24, Phidgets, MIT's Handyboard, and many others offer similar functionality. All of these tools take the messy details of microcontroller programming and wrap it up in an easy-to-use package. Arduino also simplifies the process of working with microcontrollers, but it offers some advantage for teachers, students, and interested amateurs over other systems:

- Inexpensive - Arduino boards are relatively inexpensive compared to other microcontroller platforms. The least expensive version of the Arduino module can be assembled by hand, and even the pre-assembled Arduino modules cost less than \$50
- Cross-platform - The Arduino Software (IDE) runs on Windows, Macintosh OSX, and Linux operating systems. Most microcontroller systems are limited to Windows.

- Simple, clear programming environment - The Arduino Software (IDE) is easy-to-use for beginners, yet flexible enough for advanced users to take advantage of as well.
- Open source and extensible software - The Arduino software is published as open source tools, available for extension by experienced programmers. The language can be expanded through C++ libraries, and people wanting to understand the technical details can make the leap from Arduino to the AVR C programming language on which it's based.
- Open source and extensible hardware - The plans of the Arduino boards are published under a Creative Commons license, so experienced circuit designers can make their own version of the module, extending it and improving it. Even relatively inexperienced users can build the breadboard version of the module in order to understand how it works and save money.

2.2.3 TECHNICAL DESCRIPTION

TABLE 2.1 PIN DESCRIPTION

Pin Category	Pin Name	Details
Power	Vin, 3.3V, 5V, GND	Vin: Input voltage to Arduino when using an external power source. 5V: Regulated power supply used to power microcontroller and other components on the board. 3.3V: 3.3V supply generated by on-board voltage regulator. Maximum current draw is 50mA. GND: ground pins.
Reset	Reset	Resets the microcontroller.
Analog Pins	A0 – A5	Used to provide analog input in the range of 0-5V
Input/Output Pins	Digital Pins 0 - 13	Can be used as input or output pins.

Serial	0(Rx), 1(Tx)	Used to receive and transmit TTL serial data
External Interrupts	2, 3	To trigger an interrupt.
PWM	3, 5, 6, 9, 11	Provides 8-bit PWM output.
SPI	10 (SS), 11 (MOSI), 12 (MISO) and 13 (SCK)	Used for SPI communication.
Inbuilt LED	13	To turn on the inbuilt LED.
TWI	A4 (SDA), A5 (SCA)	Used for TWI communication.
AREF	AREF	To provide reference voltage for input voltage.

2.2.4 OVERVIEW OF ARDUINO UNO

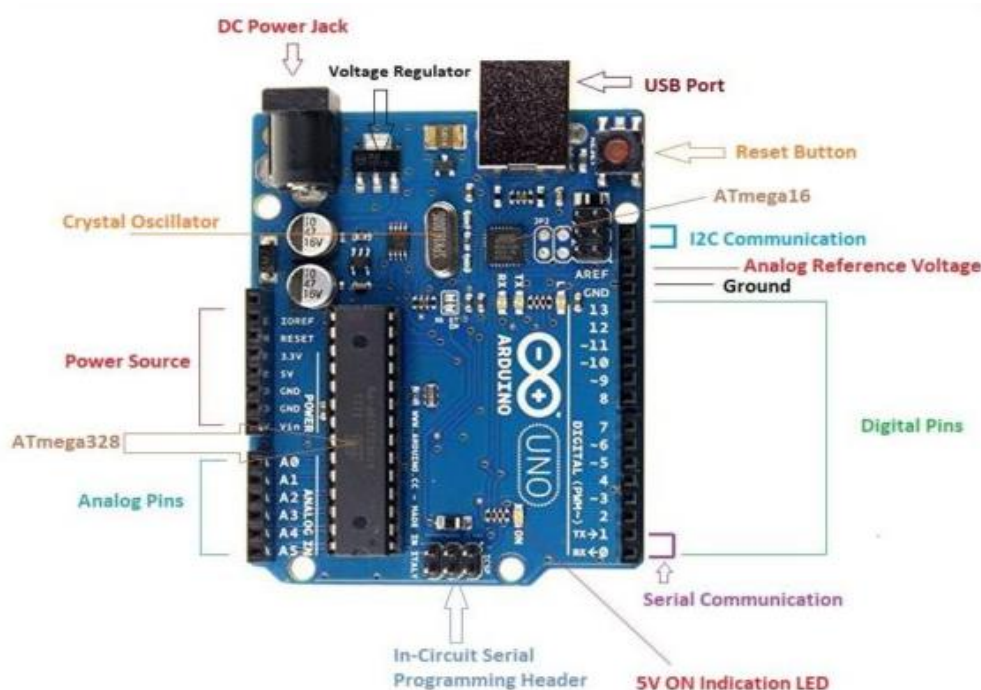


Fig 2.2 Arduino Uno Board

Arduino Uno is a microcontroller board based on 8-bit ATmega328P microcontroller. Along with ATmega328P, it consists other components such as crystal oscillator, serial communication, voltage regulator, etc. to support the microcontroller. Arduino Uno has 14

digital input/output pins (out of which 6 can be used as PWM outputs), 6 analog input pins, a USB connection, A Power barrel jack, an ICSP header and a reset button.

2.2.4.1 ARDUINO UNO TECHNICAL SPECIFICATIONS

TABLE 2.2 SPECIFICATIONS OF ARDUINO UNO

Microcontroller	ATmega328P – 8 bit AVR family microcontroller
Operating Voltage	5V
Recommended Input Voltage	7-12V
Input Voltage Limits	6-20V
Analog Input Pins	6 (A0 – A5)
Digital I/O Pins	14 (Out of which 6 provide PWM output)
DC Current on I/O Pins	40 mA
DC Current on 3.3V Pin	50 mA
Flash Memory	32 KB (0.5 KB is used for Bootloader)
SRAM	2 KB
EEPROM	1 KB
Frequency (Clock Speed)	16Hz

2.2.5 HOW TO USE ARDUINO BOARD

The 14 digital input/output pins can be used as input or output pins by using pin Mode (), digital Read () and digital Write () functions in Arduino programming. Each pin operate at 5V and can provide or receive a maximum of 40mA current, and has an internal pull-up resistor of 20-50 K Ohms which are disconnected by default. Out of these 14 pins, some pins have specific functions as listed below:

- Serial Pins 0 (Rx) and 1 (Tx): Rx and Tx pins are used to receive and transmit TTL serial data. They are connected with the corresponding ATmega328P USB to TTL serial chip.
- External Interrupt Pins 2 and 3: These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value.

- PWM Pins 3, 5, 6, 9 and 11: These pins provide an 8-bit PWM output by using analog Write() function.
- SPI Pins 10 (SS), 11 (MOSI), 12 (MISO) and 13 (SCK): These pins are used for SPI communication.
- In-built LED Pin 13: This pin is connected with an built-in LED, when pin 13 is HIGH – LED is on and when pin 13 is LOW, its off.

Along with 14 Digital pins, there are 6 analog input pins, each of which provides 10 bits of resolution, i.e. 1024 different values. They measure from 0 to 5 volts but this limit can be increased by using AREF pin with analog Reference () function.

- Analog pin 4 (SDA) and pin 5 (SCA) also used for TWI communication using Wire library. Arduino Uno has a couple of other pins as explained below:
- AREF: Used to provide reference voltage for analog inputs with analogReference() function.
- Reset Pin: Making this pin LOW, resets the microcontroller.

2.2.6 COMMUNICATION

Arduino can be used to communicate with a computer, another Arduino board or other microcontrollers. The ATmega328P microcontroller provides UART TTL (5V) serial communication which can be done using digital pin 0 (Rx) and digital pin 1 (Tx). An ATmega16U2 on the board channels this serial communication over USB and appears as a virtual com port to software on the computer. The ATmega16U2 firmware uses the standard USB COM drivers, and no external driver is needed. The Arduino software includes a serial monitor which allows simple textual data to be sent to and from the Arduino board. There are two RX and TX LEDs on the Arduino board which will flash when data is being transmitted via the USB-to-serial chip and USB connection to the computer. When ATmega328 chip is used in place of Arduino Uno, or vice versa, the image below shows the pin mapping between the two.

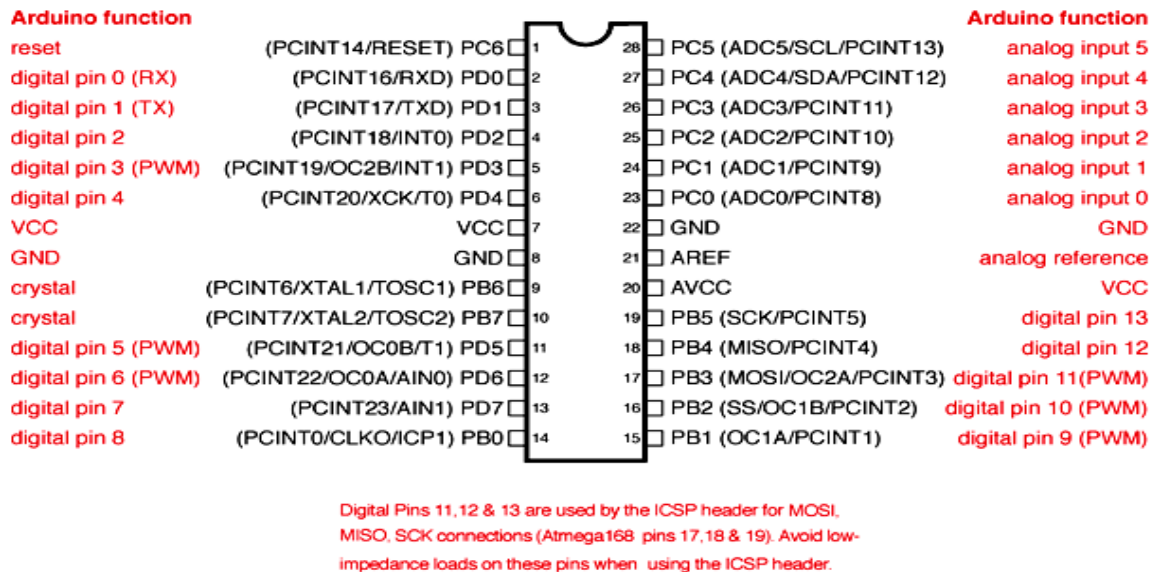


Fig 2.3 Arduino UNO ATmega328 Pin mapping

2.2.7 ADVANTAGES OF ARDUINO

- Inexpensive
- Cross-platform
- Simple, clear programming environment
- Open source and extensible software

2.2.8 APPLICATIONS OF ARDUINO UNO

- Automation.
- Robotics.
- Control Systems.
- Embedded Computer.
- Instrumentation.

2.3 VOLTAGE REGULATORS

2.3.1 7805 PIN AND CIRCUIT DIAGRAM

7805 is an easy to use voltage regulator IC which outputs 5 volts rated at 1A max. It takes an unregulated DC voltage input which can be fluctuating within its input limits and converts this fluctuating voltage input into a perfectly regulated 5 volts power output. For example, a 12 volt lead acid battery when fully charged gives out approximately 12.70 volts and when fully discharged, it gives out 10.50 volts. This difference can be even more under load or under charging state. If we use this battery as an input source for our 7805, output voltage will remain 5 regardless of that voltage difference of battery during charging and discharging phases.

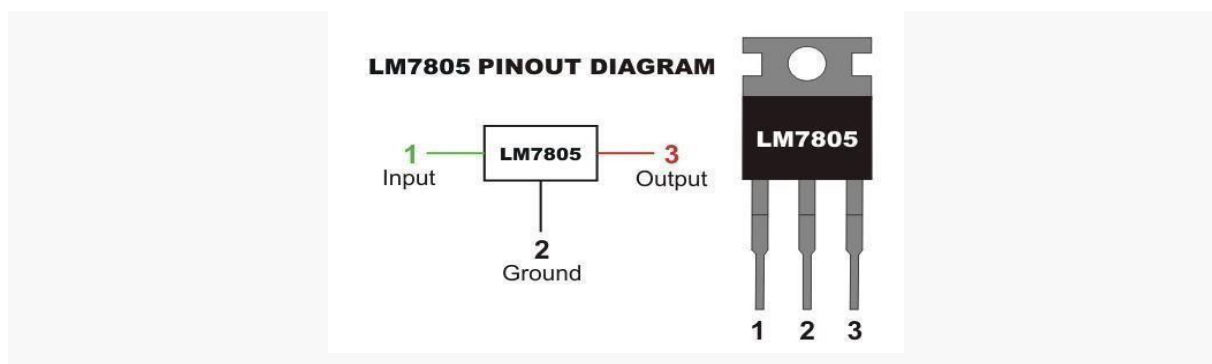


Fig 2.4 7805 Pins Diagram

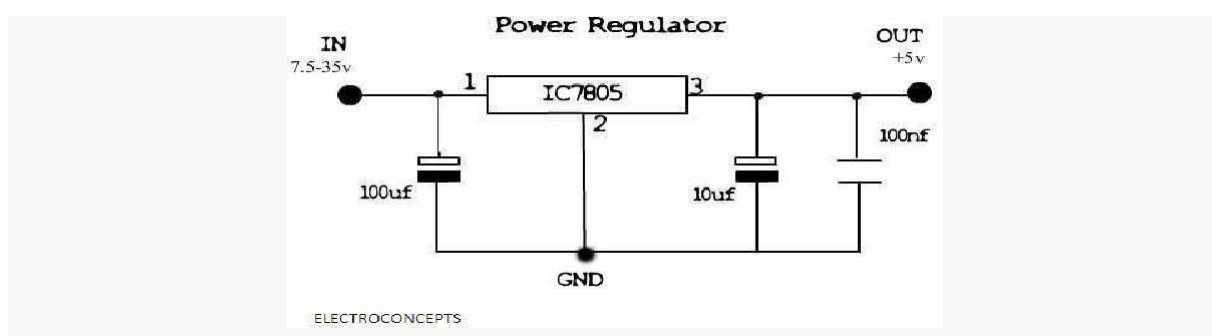


Fig 2.5 7805 Circuit Diagram

7805 IC has a thermal shutdown feature to protect the IC in case of overheating so it should be safe to use 7805 without a heatsink plate for less than 200mA load.

2.3.2 ADVANTAGES

- 7805 voltage regulator IC does not require any component to balance or saturate their output voltage.
- The 7805 IC has a built-in protection from the high current. There is a heat-sink with the common ground connected with which is helpful in order to prevent our regulator IC from overheating and short-circuits making it uncompromising in the most application.

2.3.3 7809 PIN AND CIRCUIT DIAGRAM

7809 is a voltage regulator integrated circuit (IC) which is widely used in electronic circuits. Apparently, it looks like a transistor. It has three pins. For a better understanding, I have given an image of 7809 below. You can easily see the V in and V out pins as well as the ground pin. It is really easy to use 7809 for voltage regulation purposes. I have also included a circuit diagram of 7809 so that you may learn how to use it in a circuit diagram.



Fig 2.6 7809 Pins diagram

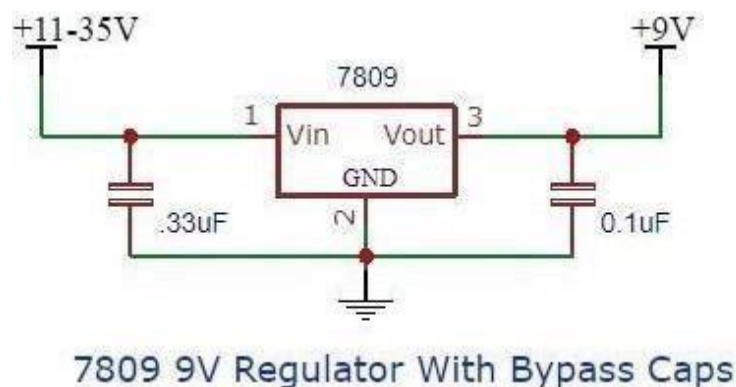


Fig 2.7 7809 Circuit Diagram

It is wise to use two 1uF capacitors on both input and output sides to filter any ripple or distortion in voltage but it is not necessary. In the image, you can see that 12V are being supplied on the input side of 7809 but the output side of 7809 is outputting Regulated 9V. As long as the input voltage remains above 9V, output voltage of 7809 will remain smooth and regulated.

Please note that input voltage of 7809 can be up to 23V but under my experience, it is wise to avoid input over 15V. 7809 is claimed to output 9V and almost 1.5A Current but again, I have experienced that we should not put a load over 9V and 1A on it. Since we are using it in power supply, the transfer of power will result in heat output. We will need to use a heat sink with 7809 otherwise this heat can damage it. It is advised to use a 1A fuse on the output side of 7809 and a 1.5A fuse on the input side of 7809 to avoid damage in case of short circuit.

2.4 LCD

2.4.1 WHAT IS LIQUID CRYSTAL DISPLAY

The liquid crystal display uses the property of light monitoring of liquid crystal and they do not emit the light directly. The Liquid crystal display is a flat panel display or the electronic visual display. With low information, content the LCD's are obtained in the fixed image or the arbitrary image which are displayed or hidden like present words, digits, or 7 segment display. The arbitrary images are made up of large no of small pixels and the element has larger elements.

2.4.2 LIQUID CRYSTAL DISPLAY OF 16x2

The 16×2 liquid crystal display contains two horizontal lines and they are used for compressing the space of 16 display characters. Inbuilt, the LCD has two registers which are described below.

- Command Register
- Data Register

Command Register: This register is used to insert a special command in the LCD. The command is a special set of data and it is used to give the internal command to the liquid crystal display like clear screen, move to line 1 character 1, setting the curser and etc.

Data Register: The data registers are used to enter the line in the LCD

TABLE 2.3 LCD PIN CONFIGURATION

Pin No	Pin Name	Pin Description
Pin 1	GND	This pin is a ground pin and the LCD is connected to the Ground
Pin 2	VCC	The VCC pin is used to supply the power to the LCD
Pin 3	VEE	This pin is used for adjusting the contrast of the LCD by connecting the variable resistor in between the VCC & Ground.
Pin 4	RS	The RS is known as register select and it selects the Command/Data register. To select the command register the RS should be equal to zero. To select the Data register the RS should be equal to one.
Pin 5	R/W	This pin is used to select the operations of Read/Write. To perform the write operations the R/W should be equal to zero. To perform the read operations the R/W should be equal to one.
Pin 6	EN	This is a enable signal pin if the positive pulses are passing through a pin, then the pin function as a read/write pin.
Pin 7	DB0 to DB7	The pin 7 contains total 8 pins which are used as a Data pin of LCD.
Pin 15	LED +	This pin is connected to VCC and it is used for the pin 16 to set up the glow of backlight of LCD.
Pin 16	LED –	This pin is connected to Ground and it is used for the pin 15 to set up the glow of backlight of the LCD.

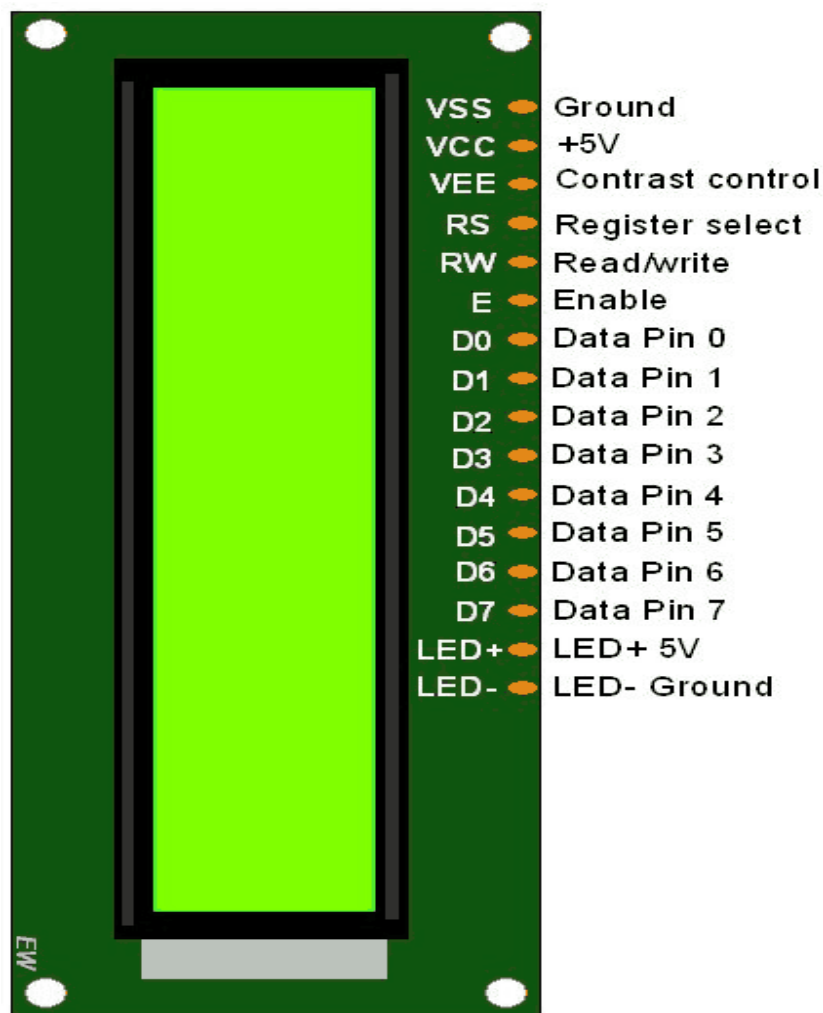


Fig 2.8 LCD Pin configuration

2.4.3 LCD INTERFACING WITH THE ARDUINO MODULE

The digital input lines (DB4-DB7) are interfaced with the Arduino pins from 2-5. To adjust the contrast of the display here we can use 1K potentiometer. The current through the back LED light is fed through 100-ohm resistor. The external power source of +5V is fed to the pin 2 of LCD and Arduino board is powered through 9V regulator. The Arduino board is built in with internal regulator of 5v such that input supply source to this board must be around 9v.

In recent years the LCD panels became very popular because of their widespread use in various electronic systems like instruments to read the parameter values, digital communications for sending or receiving the text information, data acquisition systems, etc. These display units dominating seven segment displays by providing more features to the user. The LCD system can display numbers, characters, and graphics, where as seven

segments LED's displays only numbers, there fore most of the engineers prefers LCD's.

The data fed to the LCD remains as it is and the same will be displayed until it gets an erase signal from the controller. The data can be stored and it can be refreshed for the next task.

The instruction command codes from microcontroller can be sent to the LCD to clear the display, depending up on the command the cursor can be brought to home position or blink the cursor. The LCD is having two important resistors internally, command resistor and data register; RS pin is used to select either command register or data register. If RS = 0, the instruction command code register is selected and allowing the user to send a command to clear the display. If RS = 1 the data register is selected, there by the user is allowed to send data that is to be displayed on LCD screen. By making RS pin to zero, we can also check the busy flag bit to see if the LCD is ready to receive information. As already mentioned that D0 – D7 of LCD pins are 8 – bit data pins and the busy flag is D7, it can be read when R/W (Read or Write) pin is high (R/W = 0) and RS = 0, as follows; if R/W = 1, R/S = 1. When D7 pin is high, the LCD is busy taking care of internal operations and will not accept any new information. When D7 = 0, the LCD is ready to receive new information. It is recommended to check the busy flag before writing any data to the LCD. The following is the table shows the list of instruction command codes.

TABLE 2.4 LIST OF INSTRUCTION COMMAND CODES

Code	Command to LCD Instruction
1	Clear display screen
2	Return home
4	Decrement cursor (shift cursor to left)
6	Increment cursor (shift cursor to right)
5	Shift display right
7	Shift display left
8	Display off, cursor off
A	Display off, cursor on
C	Display on, cursor off
E	Display on, cursor blinking
F	Display on, cursor blinking
10	Shift cursor position to left
14	Shift cursor position to right
18	Shift the entire display to the left
IC	Shift the entire display to the right
80	Force cursor to beginning of first line
CO	Force cursor to beginning of second line
38	2 Lines and 5x7 Matrix

To send any commands from instruction command code table to the LCD, make RS pin to zero. For data, feed high signal to RS pin, then send a high – to – low pulse to the E pin to enable the internal latch of the LCD. For this, the suitable program is to be prepared for LCD connections. Another suitable program is to be prepared for sending code to the LCD with checking busy flag. Depending up on the program the busy flag can be D7 of the command resistor, to read command register R/W pin must be high and RS pin must be low, and a low to high pulse for the enable pin will provide the command register. After reading the command register, if bit D7 (busy flag) is high, the LCD is busy and no information (either command or data) should be issued to it. During D7 is zero; at that time we can send data or commands to the LCD. In this method time delays are not required in the program, because we are checking the busy flag before issuing commands to the LCD. Enable line must be negative-edge triggered for the write and it should be positive-edge triggered for the read.

2.5 LM35 TEMPERATURE SENSOR

2.5.1 DESCRIPTION

In general, a temperature sensor is a device which is designed specifically to measure the hotness or coldness of an object. LM35 is a precision IC temperature sensor with its output proportional to the temperature (in °C) With LM35, the temperature can be measured more accurately than with a thermister. It also possesses low self heating and does not cause more than 0.1 °C temperature rise in still air. The operating temperature range is from -55°C to 150°C. The LM35's low output impedance, linear output, and precise inherent calibration make interfacing to readout or control circuitry especially easy. It has its applications on power supplies, battery management, appliances, etc.

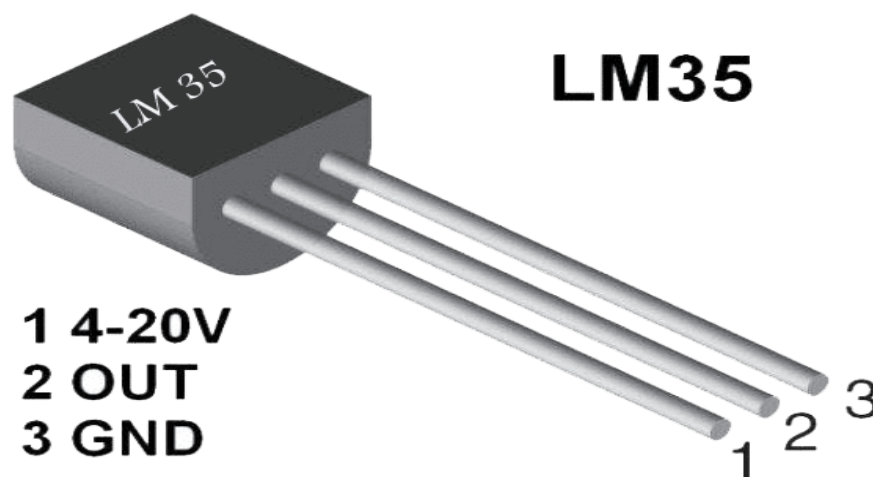


Fig 2.9 LM35 Temperature sensor pins out

The LM35 is an integrated circuit sensor that can be used to measure temperature with an electrical output proportional to the temperature (in °C). It can measure temperature more accurately. The sensor circuitry is sealed and not subject to oxidation. The LM35 generates a higher output voltage than thermocouples and may not require that the output voltage be amplified. The LM35 has an output voltage that is proportional to the Celsius temperature. The scale factor is $.01\text{V}/^{\circ}\text{C}$.

The LM35 does not require any external calibration or trimming and maintains an accuracy of $\pm 0.4^{\circ}\text{C}$ at room temperature and $\pm 0.8^{\circ}\text{C}$ over a range of 0°C to $+100^{\circ}\text{C}$. Another important characteristic of the LM35 is that it draws only 60 micro amps from its supply and possesses a low self-heating capability.

2.5.2 WORKING PRINCIPLE OF LM35

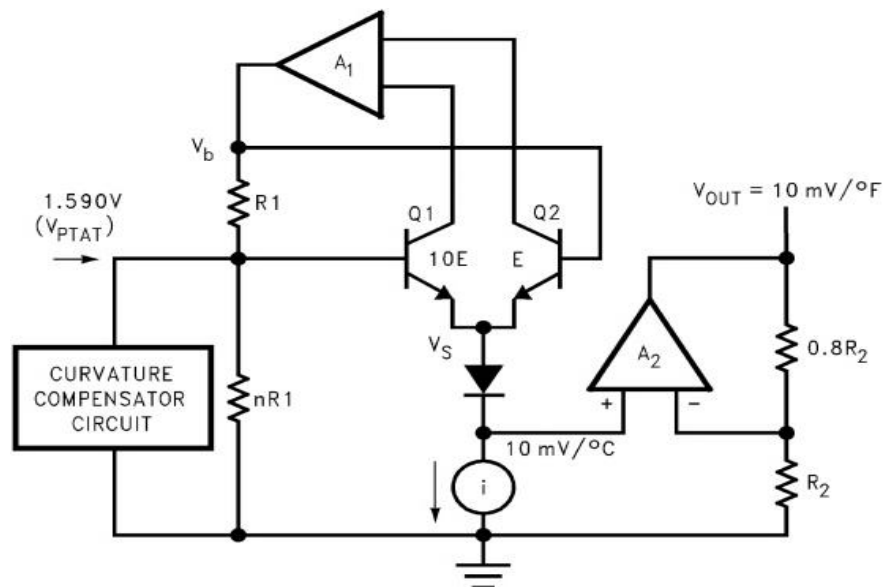


Fig 2.10 Circuit diagram of LM35

There are two transistors in the center of the drawing. One has ten times the emitter area of the other. This means it has one tenth of the current density, since the same current is going through both transistors. This causes a voltage across the resistor R1 that is proportional to the absolute temperature, and is almost linear across the range. The "almost" part is taken care of by a special circuit that straightens out the slightly curved graph of voltage versus temperature. The amplifier at the top ensures that the voltage at the base of the left transistor (Q1) is proportional to absolute temperature (PTAT) by comparing the output of the two transistors. The amplifier at the right converts absolute temperature (measured in Kelvin) into either Fahrenheit or Celsius, depending on the part (LM34 or LM35). The little circle with the "i" in it is a constant current source circuit.

The two resistors are calibrated in the factory to produce a highly accurate temperature sensor. The integrated circuit has many transistors in it -- two in the middle, some in each amplifier, some in the constant current source, and some in the curvature compensation circuit. All of that is fit into the tiny package with three leads.

The LM35 IC has 3 pins, out of 2 for the power supply and one for the analog output. It is a low voltage IC which uses approximately +5VDC of power. The output pin provides an analog voltage output that is linearly proportional to the Celsius (centigrade) temperature. Pin 2 gives an output of 1 milli-volt per 0.1°C (10mV per degree). So to get the degree value in Celsius, all that must be done is to take the voltage output and divide it by 10-this give out the value degrees in Celsius.

2.6 WIFI MODULE

2.6.1 ESP8266 WIFI MODULE

The ESP8266 is a small Wi-Fi module built around the ESP8266 chip that can connect your microcontroller to the internet wirelessly for a very small cost. It can be a great option for Internet of Things (IOT) projects, but can be difficult to work with for beginner hobbyists who do not have prior experience with the module. In this tutorial, we hope to show you how to interface the ESP8266 with an Arduino and perform some basic functions like connecting it to a Wi-Fi network.

This module has a powerful enough on-board processing and storage capability that allows it to be integrated with the sensors and other application specific devices through its GPIOs with minimal development up-front and minimal loading during runtime. Its high degree of on-chip integration allows for minimal external circuitry, including the front-end module, is designed to occupy minimal PCB area. The ESP8266 supports APSD for VoIP applications and Bluetooth co-existence interfaces.

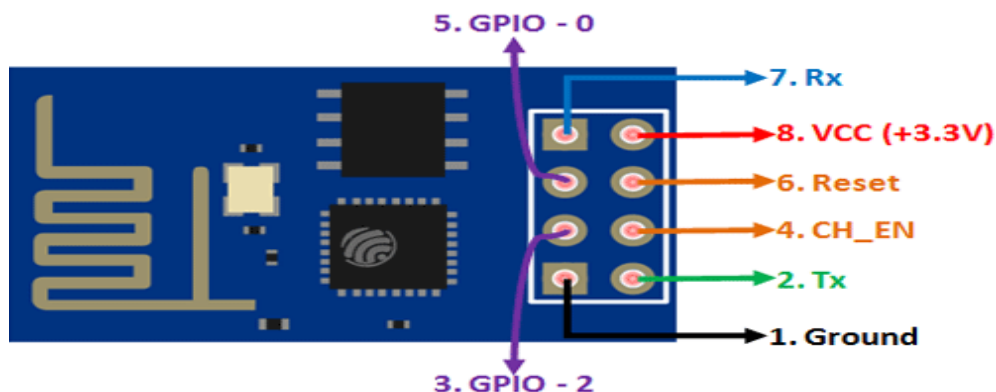


Fig 2.11 ESP8266 WI-FI module pin diagram

2.6.2 FEATURES

- Wi-Fi Direct (P2P), soft-AP
- Integrated TCP/IP protocol stack
- Integrated TR switch, balun, LNA, power amplifier and matching network
- Integrated PLLs, regulators, DCXO and power management units
- +19.5dBm output power in 802.11b mode
- Power down leakage current of <10uA
- Integrated low power 32-bit CPU could be used as application processor
- SDIO 1.1 / 2.0, SPI, UART
- STBC, 1×1 MIMO, 2×1 MIMO
- A-MPDU & A-MSDU aggregation & 0.4ms guard interval
- Wake up and transmit packets in < 2ms
- Standby power consumption of < 1.0mW (DTIM3)
- Default baud rate: 115200

2.7 POWER MOSFETS

2.7.1 POWER MOSFETS DESCRIPTION

One kind of MOSFET which handles high levels of power is known as Power MOSFET. As compared to normal MOSFETs in the less voltage range, these MOSFETs work much better by exhibiting high speed of switching. Its operating principle is the same as general MOSFETs. The most widely used power MOSFETs are p-channel Enhancement-mode, n-channel Enhancement-mode or n-channel depletion mode & p-channel depletion mode. The power MOSFET frequency is high like up to 100 kHz. The **power MOSFET symbol** is shown below.

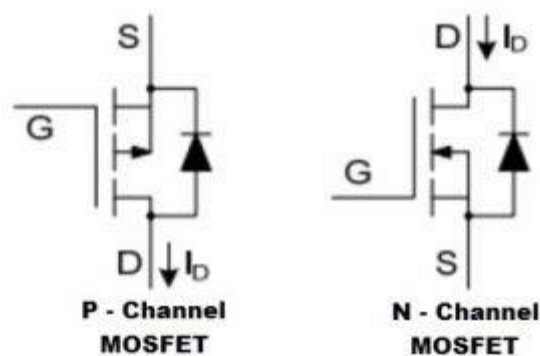


Fig 2.12 POWER MOSFET symbols

2.7.2 POWER MOSFET CIRCUIT

The power MOSFET circuit is shown below. In this circuit, the main terminals of this circuit are the source and drain. The flow of current direction will be from drain terminal to source and it is controlled through the zero voltage from gate terminal to the source. At the drain, a positive voltage is relative to the source. So it will consequence in a current of up to possibly a few 100 volts will be blocked.

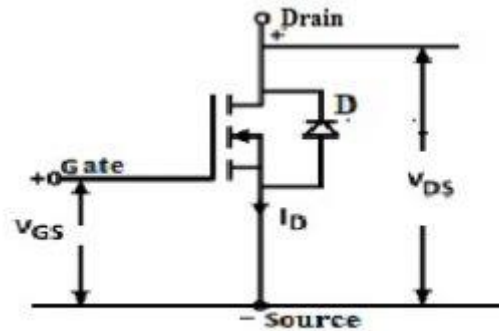


Fig 2.13 Power MOSFET circuit diagram

If a positive voltage like 3V is applied to the gate terminal then a negative charge can be induced over the surface of silicon under the gate terminal. So P-layer will become an induced N layer and allows the charge carriers like electrons to flow through it. Thus, a positive gate voltage sets up a surface channel for the flow of current from the drain terminal to the source. Here, the voltage at the gate terminal will decide the induced channel depth and in this way, the flow of current can be determined.

2.7.3 ADVANTAGES

- Very simple Gate driving circuit
- Very simple to switch ON & OFF
- It uses a high switching frequency to operate
- Thermal stability is good due to positive temperature coefficient
- Less on-state resistance
- Less expensive & Small size
- Switching speed is fast
- It is a voltage-controlled device
- Needs small power to hold it within the activated condition.

2.8 RELAY

2.8.1 FUNCTION OF RELAY

The function of relay used here is to break or make supply to the battery for charging it and disconnect the charger when temperature exceeds. A relay is a simple electromechanical switch made up of an electromagnet and a set of contacts. Relays are found hidden in all sorts of devices. Relays are simple switches which are operated both electrically and mechanically. Relays consist of electromagnet and also a set of contacts. The switching mechanism is carried out with the help of the electromagnet.

2.8.2 OPERATION OF RELAY

The main operation of a relay comes in places where only a low-power signal can be used to control a circuit. It is also used in places where only one signal can be used to control a lot of circuits. It is an electro-magnetic relay with a wire coil, surrounded by an iron core. A path of very low reluctance for the magnetic flux is provided for the movable armature and also the switch point contacts. The movable armature is connected to the yoke which is mechanically connected to the switch point contacts. These parts are safely held with the help of a spring. The spring is used so as to produce an air gap in the circuit when the relay becomes de-energized.

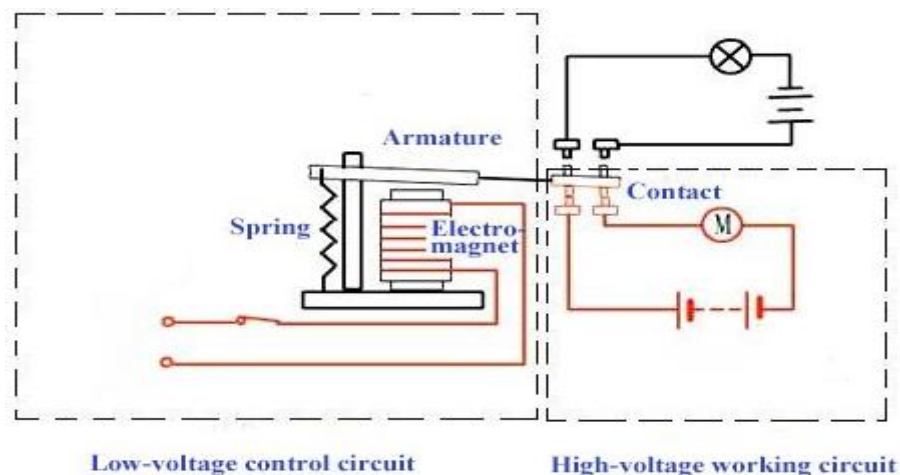


Fig 2.14 Electromagnetic Relay

2.8.3 APPLICATIONS OF RELAY

They played an important role in switching calls in telephone exchanges. They were also used in long distance telegraphy. They were used to switch the signal coming from one source to another destination. After the invention of computers they were also used to perform Boolean and other logical operations.

2.9 RESISTORS

A resistor is a passive two-terminal electrical component that implements electrical resistance as a circuit element.

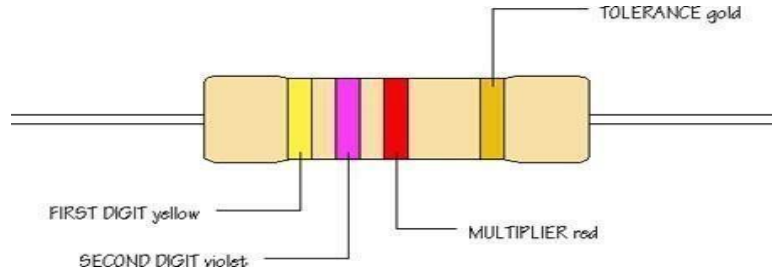


Fig 2.15 Resistor

The current through a resistor is in direct proportion to the voltage across the resistor's terminals. This relationship is represented by Ohm's law:

$$I = \frac{V}{R}$$

where I is the current through the conductor in units of amperes, V is the potential difference measured across the conductor in units of volts, and R is the resistance of the conductor in units of ohms (symbol: Ω). The ratio of the voltage applied across a resistor's terminals to the intensity of current in the circuit is called its resistance, and this can be assumed to be a constant (independent of the voltage) for ordinary resistors working within their ratings.

Resistors are common elements of electrical networks and electronic circuits and are ubiquitous in electronic equipment. Practical resistors can be made of various compounds and films, as well as resistance. Resistors are also implemented within integrated circuits, particularly analog devices, and can also be integrated into hybrid and printed circuits. The electrical functionality of a resistor is specified by its resistance: common commercial resistors are manufactured over a range of more than nine orders of magnitude. The temperature coefficient of the resistance may also be of concern in some precision applications. Practical resistors are also specified as having a maximum power rating which must exceed the anticipated power dissipation of that resistor in a particular circuit: this is mainly of concern in power electronics applications.

2.10 BUZZER

An audio signaling device like a beeper or buzzer may be electromechanical or piezoelectric or mechanical type. The main function of this is to convert the signal from audio to sound. Generally, it is powered through DC voltage and used in timers, alarm devices, printers, alarms, computers, etc. Based on the various designs, it can generate different sounds like alarm, music, bell & siren.



Fig 2.16 Buzzer

It includes two pins namely positive and negative. The positive terminal of this is represented with the '+' symbol or a longer terminal. This terminal is powered through 6Volts whereas the negative terminal is represented with the '-' symbol or short terminal and it is connected to the GND terminal.

2.11 CONNECTING WIRES

Connecting wires are used to connect the components.

2.12 CONCLUSION

In this chapter, we made a brief discussion on components of project.

CHAPTER – 3

OPERATION

3.1 INTRODUCTION

In this chapter, we will discuss about the operation of the project.

3.2 BLOCK DIAGRAM OF HI-TECH BATTERY CHARGER

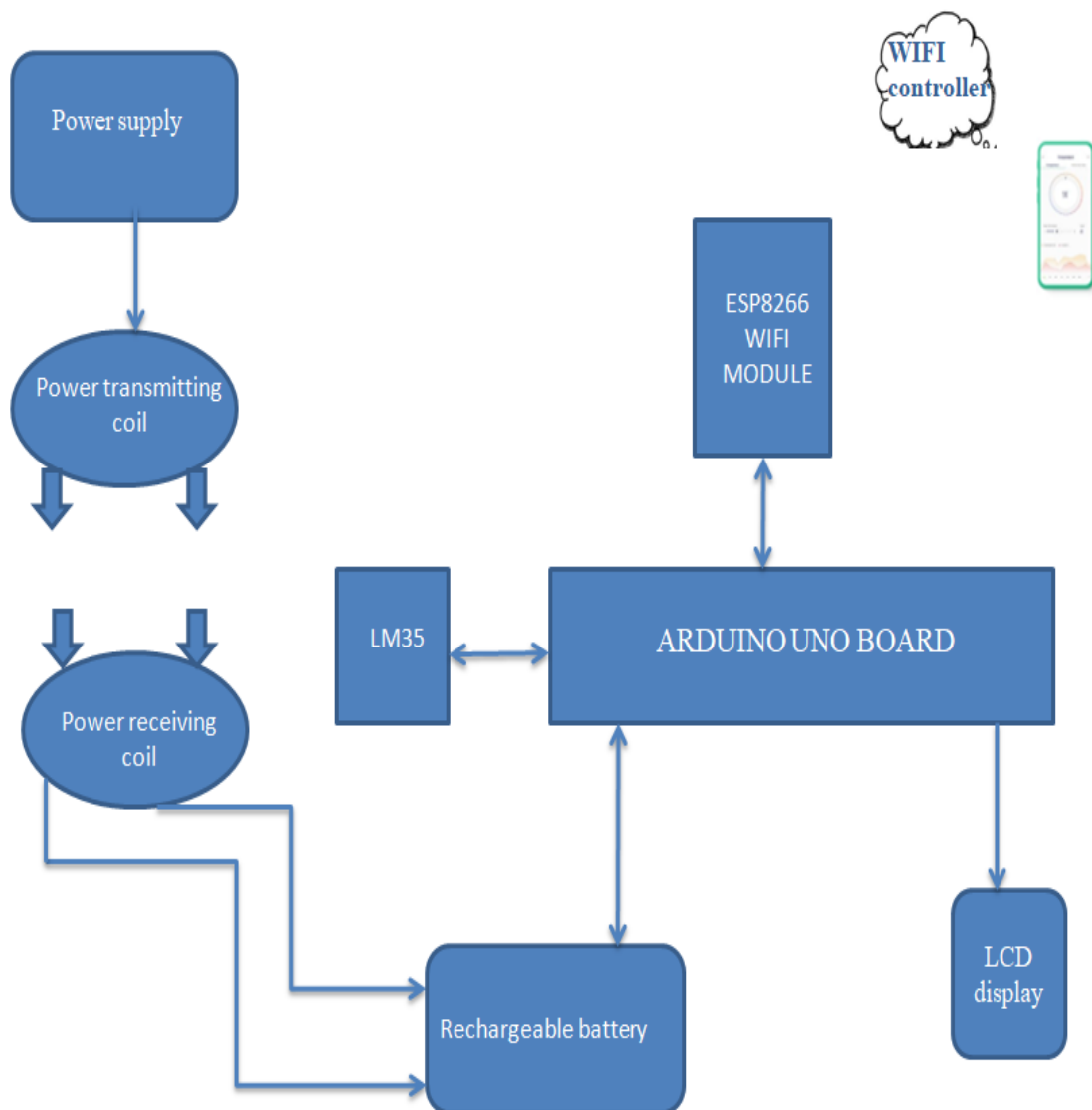


Fig 3.1 Block Diagram

3.3 CIRCUIT DIAGRAM

As per the block and circuit diagrams shown in this chapter, the process or functional description begins with the Remote control unit. Since this project work proves basic theory of wireless energy source by incorporating Electromagnetic field coils, little energy will be transmitted to a distance of nearly 35 cms.

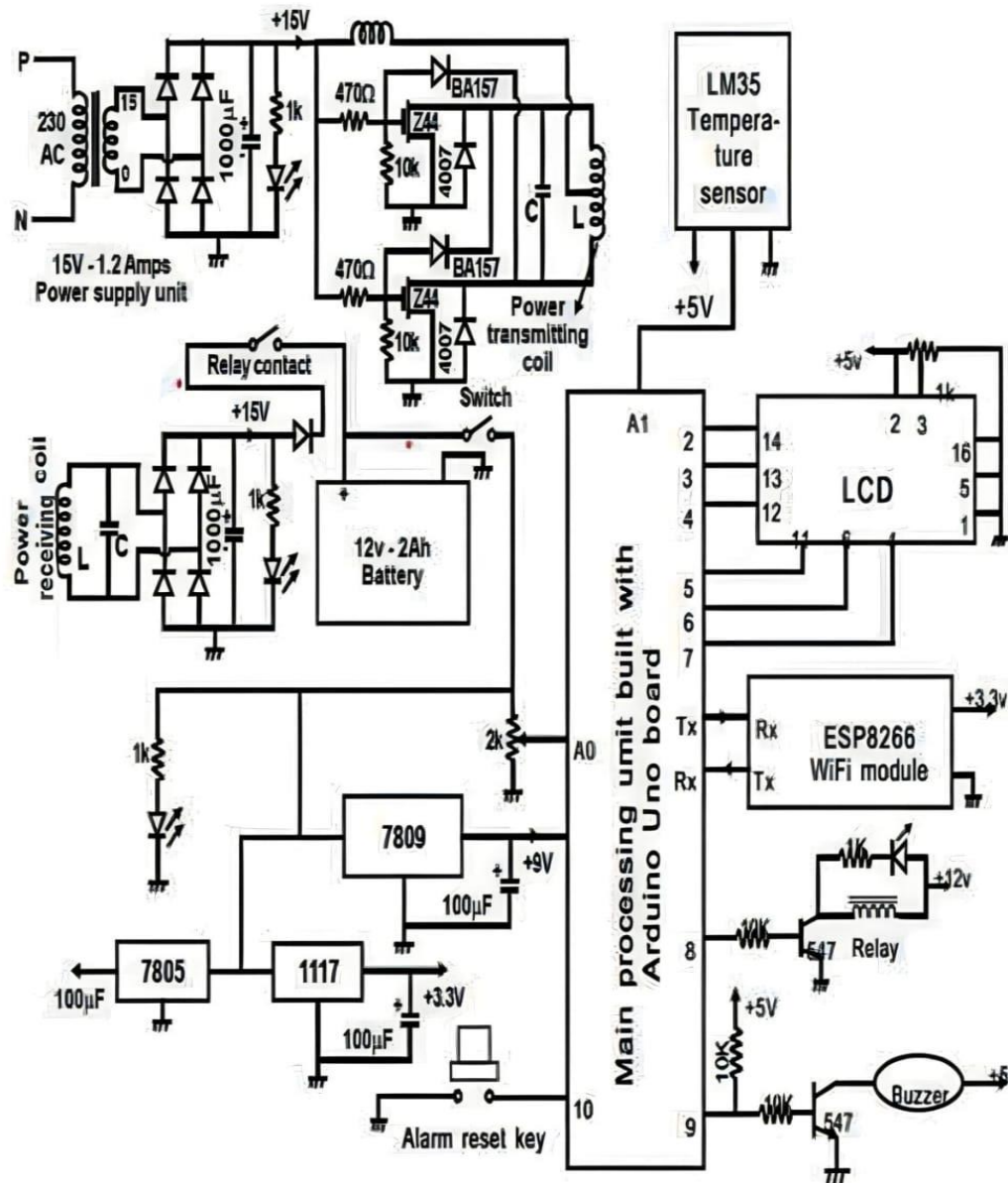


Fig 3.2 Circuit diagram

During our trial runs, we found that around 300milli amps current is obtained at a source voltage of 12 approximately. This power is enough to charge the battery arranged in the

electric vehicle. The battery used here is rated for 12V - 2Ah (Ampere hour), when this battery is charged with 300ma power source, than the charging time can be defined as battery rating / charging current rating, I.e. $2 / 0.3 = 6.6$ hours. So if a high power source is used, then the battery can be charged in less time. The detailed functional description is provided in following chapter and the following are the important building blocks used in this system.

3.4 PCB HARDWARE DIAGRAM

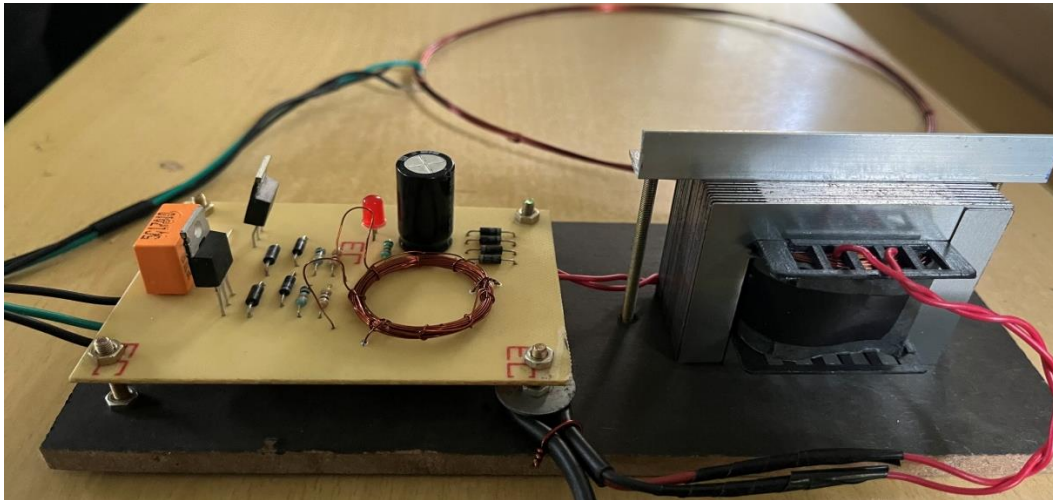


Fig 3.3 Power Transferring circuit

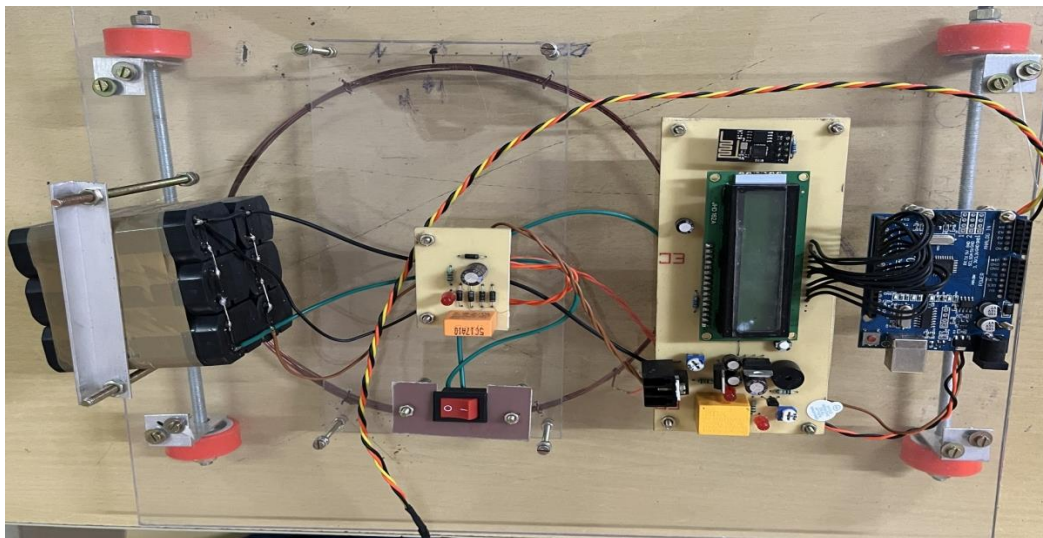


Fig 3.4 Power Receiving circuit

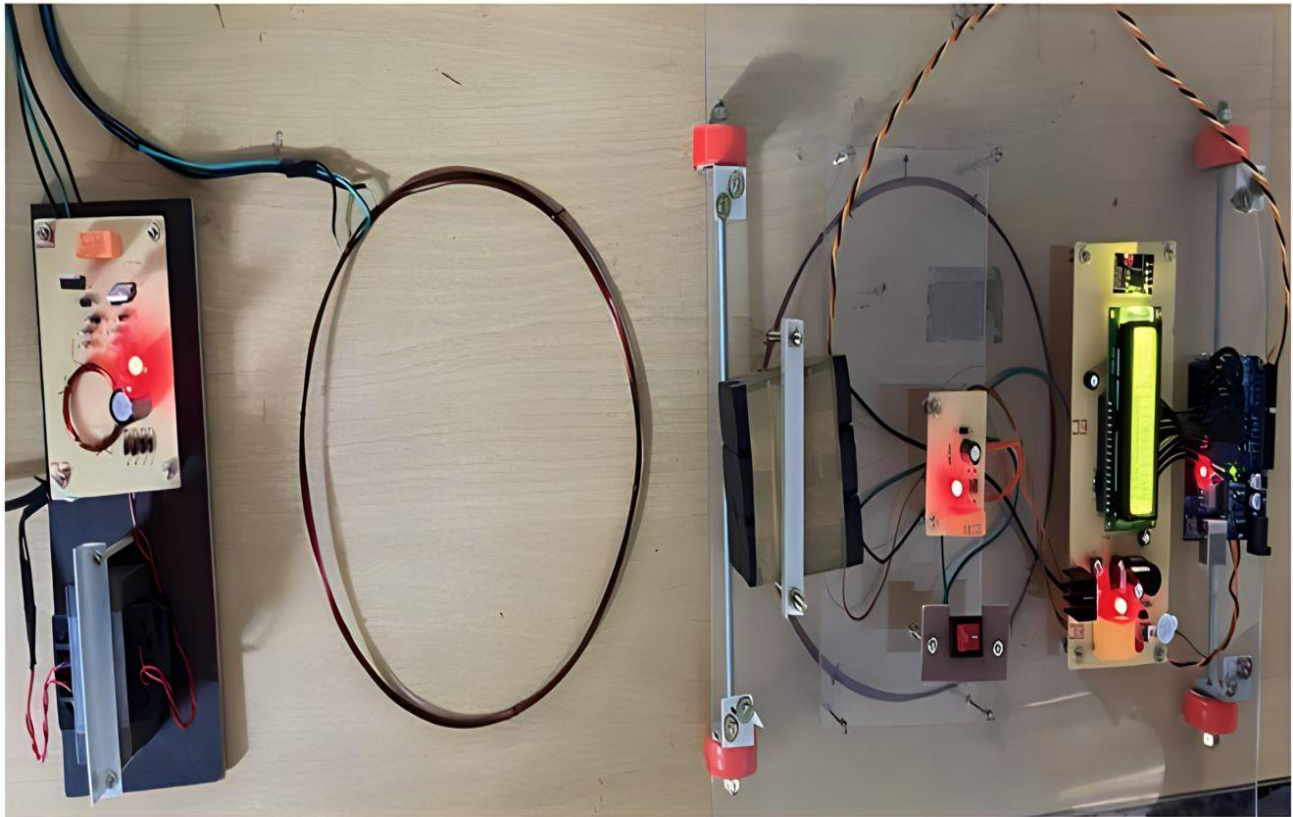


Fig 3.5 PCB hardware diagram

The PCB is designed at first level of construction, component and circuit mounted on it, and sensors are connected to it.

3.5 WORKING

Design Implementation of Cost Effective Wireless Power Transmission system is presented in this project work, since it is a prototype module low power transmitter circuit is constructed with power MOSFETs. To induce more current in to the primary coil, two MOSFETs are used and are configured in Push-pull mode of operation. With the help of a diode connected in feedback mode to both power MOSFETs and switches in a sequence (alternatively one after another) such that both MOSFETs will not conduct at a time. As the primary coil is made as centre tapped and it is divided in to two sections, each section will be energized individually through corresponding MOSFET. If top MOSFET energized bottom remains in de-energized condition, similarly if bottom MOSFET energized top remains in off condition. In this manner of switching, the switching frequency depends up on the fast recovery diodes connected in feedback loop.

3.5.1 LC CIRCUIT DESCRIPTION

An LC circuit, oscillating at its natural resonant frequency, can store electrical energy. A capacitor stores energy in the electric field (E) between its plates, depending on the voltage across it, and an inductor stores energy in its magnetic field (B), depending on the current through it. If an inductor is connected across a charged capacitor, current will start to flow through the inductor, building up a magnetic field around it and reducing the voltage on the capacitor. Eventually all the charge on the capacitor will be gone and the voltage across it will reach zero. However, the current will continue, because inductors resist changes in current. The current will begin to charge the capacitor with a voltage of opposite polarity to its original charge. Due to Faradays law, the EMF which drives the current is caused by a decrease in the magnetic field, thus the energy required to charge the capacitor is extracted from the magnetic field. When the magnetic field is completely dissipated the current will stop and the charge will again be stored in the capacitor, with the opposite polarity as before. Then the cycle will begin again, with the current flowing in the opposite direction through the inductor.

3.5.2 POWER RECEIVING CIRCUIT DESCRIPTION

The LC circuit implemented at primary side (from where the power is transmitted), the same circuit must be adopted at secondary side. The secondary coil must be exactly similar to the primary coil and there should not be any difference in the wire gauge, no. of turns and coil size. Similarly capacitor value must be the same as primary capacitor and here high quality capacitors are preferred for proper resonance coupling. If the secondary coil is synchronized perfectly with the primary coil, distance can be increased affectively. The output of secondary coil is converted in to pure DC, with the help 4 rectifiers full wave bridge is constructed and to obtain pure DC filter capacitor is used. In our trail runs we carried so many experiments by winding different coils with different gauge magnetic wires with different sizes. All these are aimed to select an effective coil by which maximum voltage can be obtained from the secondary coil. If more voltage is obtained distance also can be increased considerably. Finally we have selected one coil which is having 8 turns at secondary and 4+4 Turns at primary. Wire thickness is considered as 21 SWG and coil ring size is 8 inch. With this coil we got nearly 18V DC at a distance of 50mm, since it is no load voltage and when a load of 30ma is connected across the DC source, the voltage is fallen down to 12V.

3.5.3 TEMPERATURE SENSING CIRCUIT

This circuit constructed with LM35 is aimed to measure the battery body temperature. The process begins with the LM35 and it is a temperature sensing device which generates proportionate voltage from its output pin according to the temperature or heat applied to the sensor. The LM35 series are precision integrated-circuit temperature devices with an output voltage linearly-proportional to the Centigrade temperature. The low-output impedance, linear output and precise inherent calibration of the LM35 device makes interfacing to readout or control circuitry especially easy.

3.5.4 BATTERY TERMINAL VOLTAGE MONITORING SYSTEM

Here a simple pot or pentameter (Variable resistor) is used to measure the battery voltage, since the battery voltage will be greater than 12v dc and even greater than this when it is in charging mode, here this voltage as it cannot be applied to the A to D pin of Arduino board. Since the processor operates at a stable supply source +5v, the voltage applied to any pin of Arduino should not exceed more than 5v and there by this voltage must be adjusted accordingly through this pot. Here reference voltage will be considered and accordingly the processor will be programmed. For example, if the battery voltage is 10 it must be divided by one, i.e. $10/1 = 1$, means the output voltage of pot must be adjusted 1 volt that is proportionate to 10v.

3.6 ANDROID APP

3.6.1 HOME PAGE

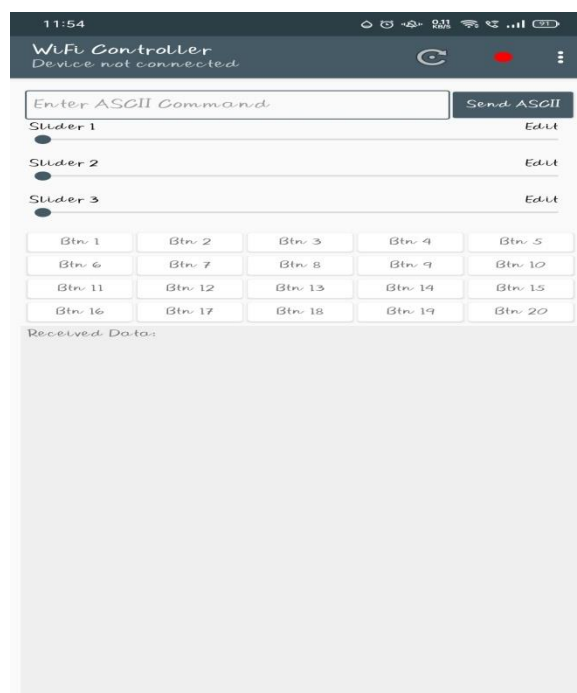


Fig 3.6 Home page display

3.6.2 LOGIN/SIGN UP PAGE

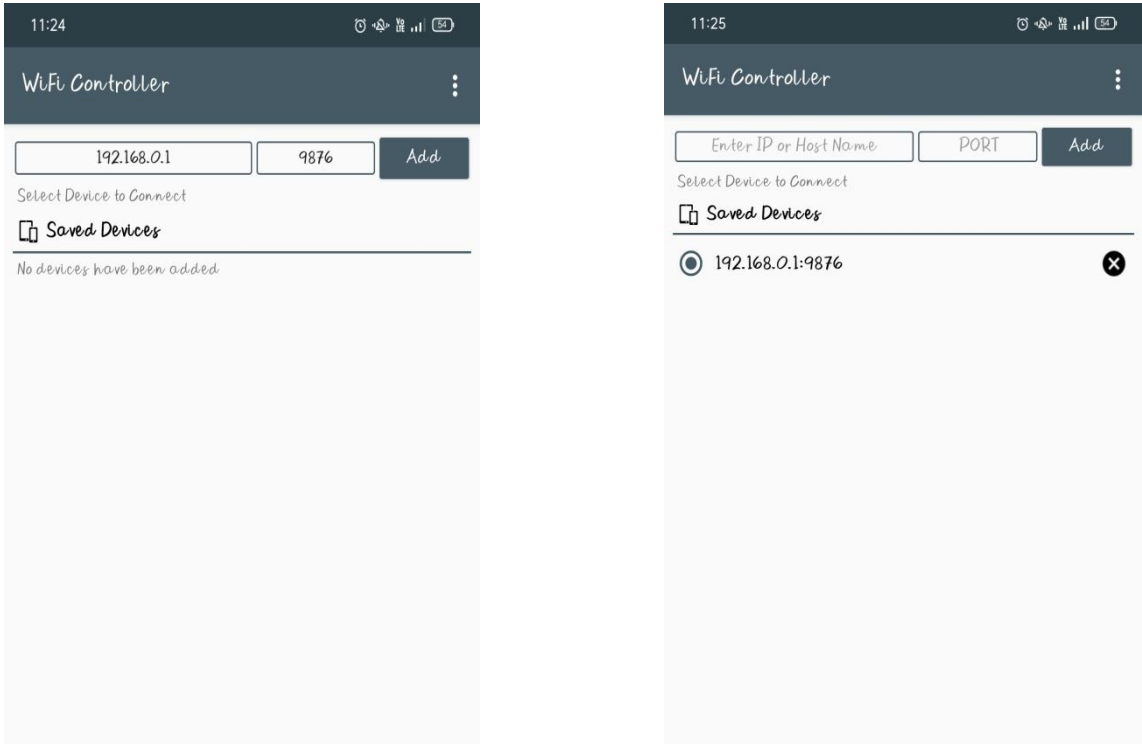


Fig 3.7 Login/sign up page

3.6.3 DASHBOARD

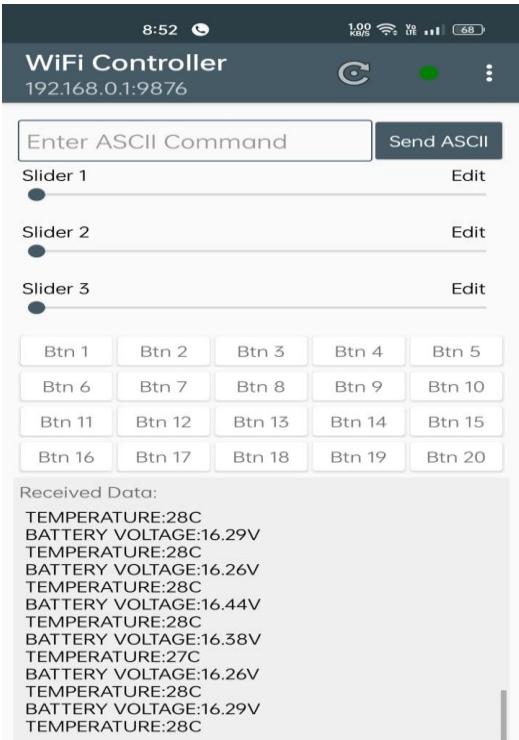


Fig 3.8 Dashboard

Through Website we can see the output. When the system get started ac current given to the kit and these ac current converts to dc current and then arduino and WIFI gets on. Website wasdesigned for the Internet of Things. It can control hardware remotely, it can display current data, it can store data, visualize it and do many other cool things. Instead, it's supporting hardware of your choice. Whether your Arduino or NodeMCU is linked to the Internet over Wi-Fi, App will get youonline and ready forthe Internet of you Things.

3.7 HARDWARE DIAGRAM OF COMPLETE CIRCUIT

The current readings from current sensors are sent to website through ESP8266. The App will display the values of current. Thus like this when the kit is located on any parking field and WIFI is provided we can observe its real time values on our android phone anywhere at any time.

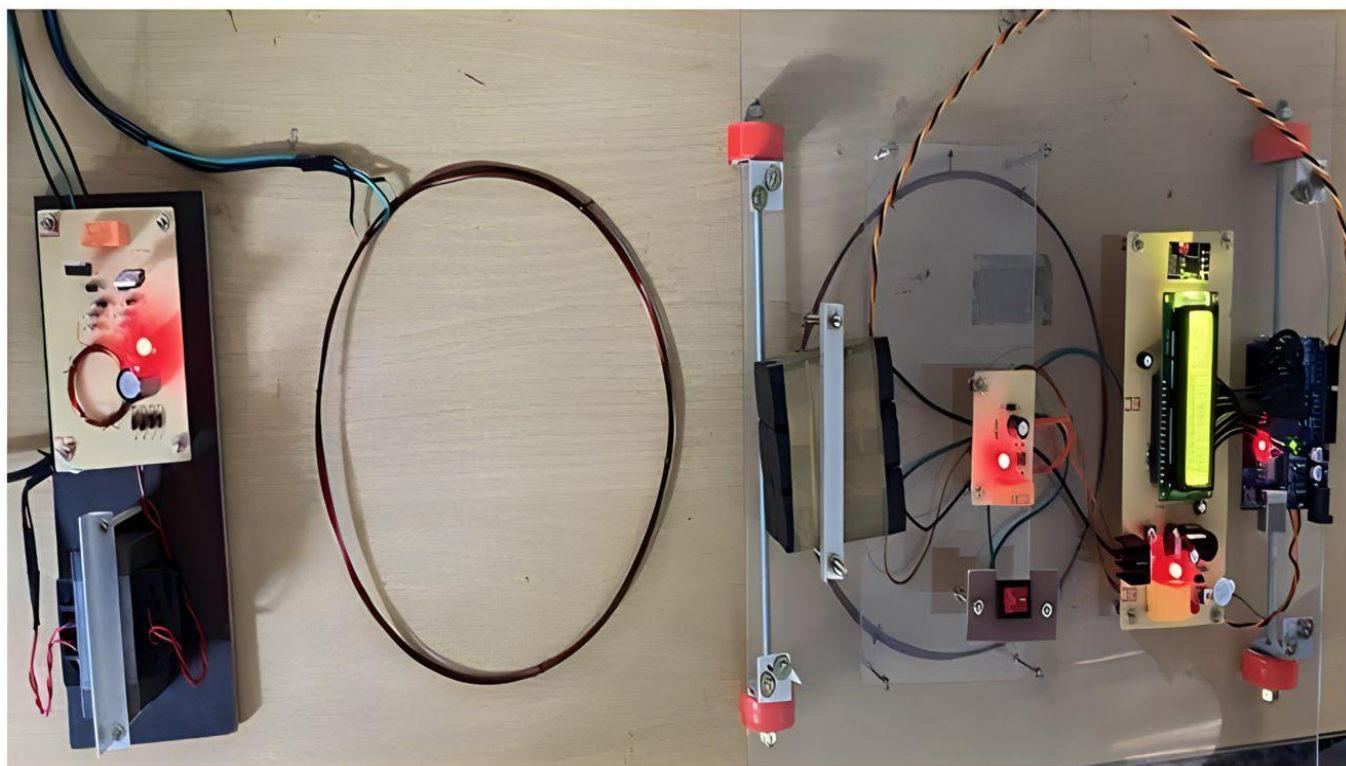


Fig 3.9 Hardware diagram of complete circuit

3.8 CONCLUSION

In this chapter, we made a brief discussion on the operation of project.

CHAPTER - 4

RESULT & DISCUSSION

4.1 INTRODUCTION

In this chapter, we will discuss about the result of the project.

4.2 RESULT & DISCUSSION

This project deals with the subject of battery protection during charging and Electric car battery charging using wireless battery charging technology is focused in this project work, in this regard function of Electro-magnetic field, Resonant inductive coupling, LC network, Synchronization between two Inductive coils, etc. are studied properly, i.e. before starting the project work. Most of the information gathered from various websites. Thermal protection concept is our own idea such that battery will be protected burning due to the overheating.

The main important factor involved in this project work is tuning of resonant circuits consisting of inductance and capacitance. The result is depended up on these two devices and another important issue is that the frequency at which rate these devices are conducting effectively. To know exactly, we came to know that formulas are not so affective, only thing is that go on with practical oriented experimented by conducting lot of trail runs with huge patience.

The resonant circuits at both sides of coils must be tuned perfectly to resonate at the same frequency. Here two similar LC circuits must be placed parallel to each other, a transmitter coil in one device transmits electric power in to space, if receiver coil in parallel to a certain device electric energy will be transmitted and received. This technology is being developed for powering and charging portable devices like cell phones, the same technology also can be used for many other applications. Here this technology is used to charge the battery.

Resonant transfer works by making a coil ring with an oscillating current. This generates an oscillating magnetic field. Because the coil is highly resonant, any energy placed in the coil dies away relatively slowly over very many cycles; but if a second coil is brought near it, the coil can pick up most of the energy before it is lost, even if it is some distance away.

4.3 CURRENT AND BATTERY VOLTAGE VALUES

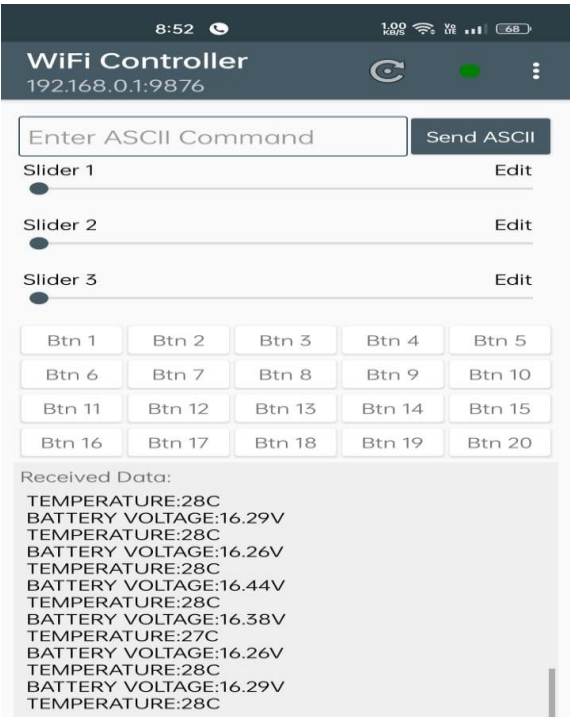


Fig 4.1 Battery voltage increases if temperature decreases

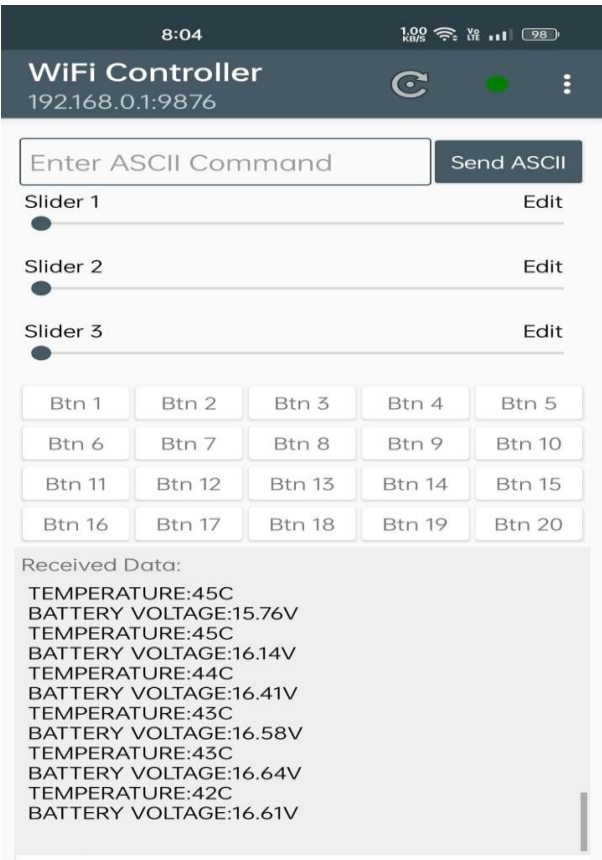


Fig 4.2 Battery voltage decreases if temperature increases

4.4 ALERT TYPE

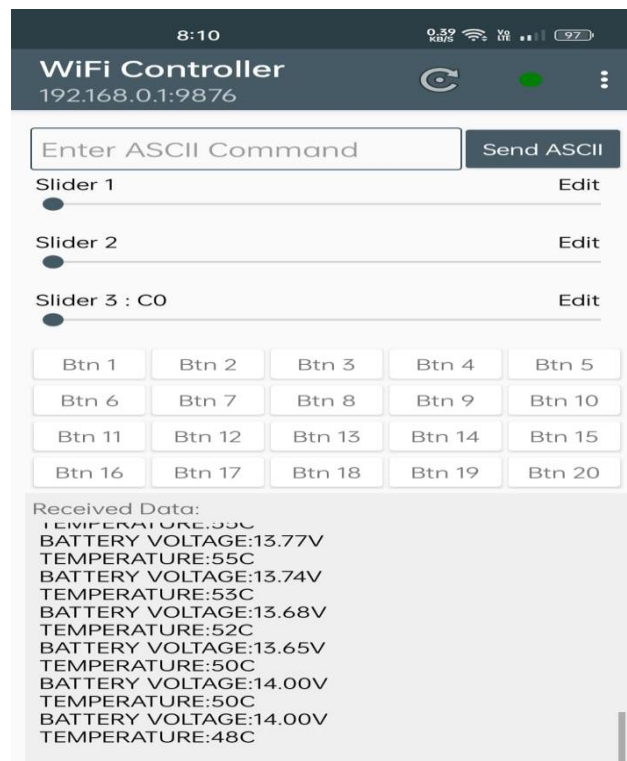


Fig 4.3 Buzzer sounds, (temp > 50°C)

4.5 CONCLUSION

In this chapter, we made a brief discussion on the result of the project.

CHAPTER - 5

CONCLUSION

5.1 INTRODUCTION

In this chapter, we will get the conclusion of Hi-tech battery charger for modern EV's using IOT

5.2 CONCLUSION

In conclusion, the development of a hi-tech battery charger for modern electric vehicles using the Internet of Things technology is a significant step towards making EVs more convenient and efficient for users. Wireless power transmission is an important subject which creates lot of enthusiasm while constructing the project work. The field of wireless transmission of electrical energy between two magnetically coupled coils requires lot of experiments to obtain better results. In our trial runs we have winded many types of magnetic coils, we have tried with different Gauge wires, with different turn's ratio, different sized coils, etc. Finally we have focused and concluded on one set of coils winded with 21 SWG wire with a ring size of 8", both primary and secondary coils are having six turns each. With these coils we found that the range increased slightly when compared with other coils. Finally the distance between the two coils is defined as 50mm, at this distance the battery is charged with less current, when the distance is decreased by less than 30mm, we observed that the battery is charging at around 300ma.

Since it is a prototype module, low power transmitter is constructed because of restricted power source at primary side. Not only power source, economy is also criteria, but after conducting so many experiments we are very confident that we can build a high power transmitter and range also can be increased accordingly.

5.3 FUTURE SCOPE:

- To reduce the battery temperature we can use intake fans to cool down the battery temperature as it can save from EV explosion.
- SOC and SOH calculation can be implemented.

5.4 CONCLUSION

In this chapter, we concluded the Hi-tech battery charger for modern EV's using IOT

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Page(s): 112 - 116
Date of Publication: 29 December 2015 - Publisher: IEEE.

APPENDIX

ARDUINO PROGRAM

```
const byte rxPin = 2; // Wire this to Tx Pin of ESP8266

const byte txPin = 3; // Wire this to Rx Pin of ESP8266

// We'll use a software serial interface to connect to ESP8266

SoftwareSerial ESP8266 (rxPin, txPin);

void setup () {

    Serial.begin(9600);

    // You may need to uncomment this block for the first run:

    // ESP8266.begin(115200); // Change this to the factory baudrate used by ESP8266

    // delay (1000);

    //

    // Serial.println("Setting BAUDRATE to 9600");

    // ESP8266.println("AT+IPR=9600");

    //

    ESP8266.begin(9600);

}

bool received = false;

void loop () {

    if (Serial.available() > 0)

    {

        String command = Serial.readStringUntil('\n');
```

```
Serial.println("Command Sent: " + command);

Serial.println();

ESP8266.println(command);

}

int responseCounter = 0;

if (ESP8266.available() > 0)

{

while (ESP8266.available() > 0)

{

if (responseCounter == 0)

{

Serial.println("Response Received:");

}

String response = ESP8266.readStringUntil('\n');

Serial.println(response);

responseCounter++;

}

Serial.println();

Serial.println("=====");

Serial.println();

}

}
```

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(7, 6, 5, 4, 3, 2);
const int RLY=8;
const int BUZ=9;
const int RST=10;
const int temp=A1;
const int vlt=A0;

float vout;
int tempc=0;
float humc=0;
int fluxc=0;
int flag=0;
void setup()
{
  lcd.begin(16, 2);
  lcd.setCursor(0,0);
  lcd.print("TEMPERATURE:  ");
  lcd.setCursor(0,1);
  lcd.print("BATT VOLT:  ");
  pinMode(temp,INPUT);
  pinMode(vlt,INPUT);
  pinMode(RST,INPUT);
  pinMode(BUZ,OUTPUT);
  pinMode(RLY,OUTPUT);
  digitalWrite(RLY,LOW);
  digitalWrite(BUZ,LOW);
  digitalWrite(RST,HIGH);
  Serial.begin(9600);
  Serial.println("WELCOME");
}
```

```
void loop()
{
  if(digitalRead(RST)==LOW)
  {
    digitalWrite(BUZ,LOW);
    flag=1;
  }
  // vout=analogRead(temp);
  vout=analogAvg1(temp);
  vout=(vout*500)/1023;
  tempc=vout;
  lcd.setCursor(13,0);
  lcd.print(tempc);
  lcd.print("C");
  Serial.print("TEMPERATURE:");
  Serial.print(tempc);
  Serial.println("C");

  if(tempc>50)
  {
    digitalWrite(RLY,HIGH);
    if(flag==0)
    {
      digitalWrite(BUZ,HIGH);
    }
  }
  else
  {
    digitalWrite(RLY,LOW);
    flag=0;
  }
  delay(300);
  vout=analogAvg(vlt);
```

```
// vout=(vout*500)/1023;
// int sensorValue = analogRead(vlt);
float voltage = vout * (5.0 / 1024.0);
voltage=(voltage*6);
lcd.setCursor(10,1);
lcd.print(voltage);
lcd.print('V');
Serial.print("BATTERY VOLTAGE:");
Serial.print(voltage);
Serial.println("V");
delay(300);

}

int analogAvg (int vlt)
{
  unsigned int total=0;
  for(int n=0; n<32; n++ )
    total += analogRead (vlt);
  return total/32;
}

int analogAvg1 (int temp)
{
  unsigned int total=0;
  for(int n=0; n<32; n++ )
    total += analogRead (temp);
  return total/32;
}
```